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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2025

ME1103-1 – THEORY OF MACHINES

(For ME)

Duration: 3 Hours

Max. Marks: 100

Note: Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which of the following disciplines provides study of relative motion between the parts of a machine
A) Theory of machines
B) Applied mechanics
C) Mechanisms
D) Kinematics
2. The example of lower pair is
A) Shaft revolving in a bearing
B) Straight line motion mechanisms
C) Automobile steering gear
D) All of the above
3. In static equilibrium of a mechanism, a member subjected three forces simulate
A) Momentum
B) A closed force polygon
C) A closed couple polygon
D) Both B and C
4. What is the use of beam engine?
A) To convert rotary motion into oscillatory motion
B) To convert rotary motion into reciprocating motion
C) To convert reciprocating motion into rotary motion
D) To convert oscillatory motion into rotary motion
5. If the link next to the shortest link is fixed, what type of mechanism is obtained?
A) Double crank mechanism
B) Linkage is not planar
C) Double rocker mechanism
D) Crank rocker mechanism
6. For what purpose are the quick return mechanisms used?
A) To convert reciprocating motion into oscillatory motion
B) To convert oscillatory motion into reciprocating motion
C) To convert reciprocating motion into rotary motion
D) To convert rotary motion into reciprocating motion
7. Watt's mechanism is based on
A) inversions of single slider chain mechanism
B) inversions of double slider chain mechanism
C) inversions of four bar chain mechanism
D) inversions of five bar chain mechanism
8. Scotch yoke mechanism is the inversion of
A) Single slider kinematic chain
B) Double slider kinematic chain
C) Four bar chain
D) None of the above
9. What is the primary purpose of a velocity polygon in the analysis of mechanisms?
A) To find the link lengths
B) To determine angular velocities of links
C) To analyze static forces
D) To draw acceleration diagrams
10. In a graphical analysis, the velocity of a point is represented by:
A) A straight line
B) A vector
C) A curve
D) A scalar quantity

11. Relative velocity is used to determine the motion of:
 A) Points in the same link B) Points in different links
 C) A fixed link only D) A point on the ground
12. In an automobile, if the vehicle makes a left turn, the gyroscopic torque
 A) increases the forces on the outer wheels B) decreases the forces on the outer wheels
 C) does not affect the forces on the outer wheels D) none of the mentioned
13. The number of teeth per unit length of the pitch circle diameter is called
 A) diametral pitch B) module
 C) clearance D) gear ratio
14. The ratio of the number of teeth on the gear to the number of teeth on the pinion is called _____
 A) velocity ratio B) module
 C) gear ratio D) pitch point
15. The radial height of a tooth above the pitch circle is called _____
 A) addendum B) dedendum
 C) rack D) line of centres
16. The radial depth of a tooth below the pitch circle is called _____
 A) dedendum B) addendum
 C) rack D) line of centres
17. For continuous transmission of motion, the contact ratio i.e. n must always be
 A) equal to 1 B) equal to 0
 C) more than unity D) less than unity
18. Frequency of oscillation of simple pendulum depends upon
 A) Length B) Acceleration due to gravity
 C) Both A & B D) Mass of the BOB
19. Which of the following is an example of a simple harmonic motion ?
 A) $x_1 = A_1 \sec \omega t$ B) $x_1 = A_1 \operatorname{cosec} \omega t$
 C) $x_1 = A_1 \cos \omega t$ D) $x_1 = A_1 \tan \omega t$
20. Fluid resistance causes damping which is known as
 A) Resistance damping B) Fluid damping
 C) Liquid damping D) Viscous damping

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

1. a) With a neat sketch explain the working of crank and lever mechanism.
 b) Explain the significance of degrees of freedom in mechanical systems.
 c) Fig.Q1(c) shows the static equilibrium of the mechanism determine the torque to be applied on link AB. Link lengths are as follows AB=20cm, BD=40cm.

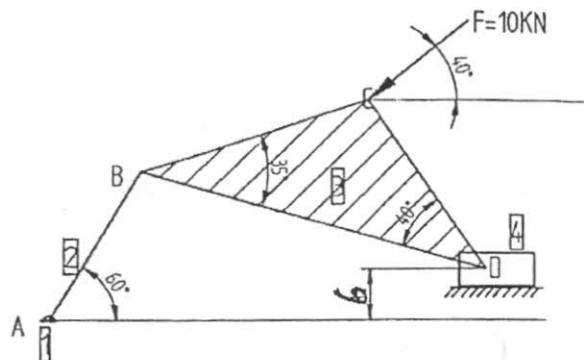


Fig.Q1(c).

Marks	BT*	CO*	PO*
04	L*1	1	1
04	L1	1	1

08	L3	2	2
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2. a) Discuss the crank and slotted lever quick return motion mechanism as an inversion of the single slider crank chain. Explain its significance in shaping machines.
- b) Determine the torque to be applied on link AB for the static equilibrium of the mechanism. Link lengths are as follows

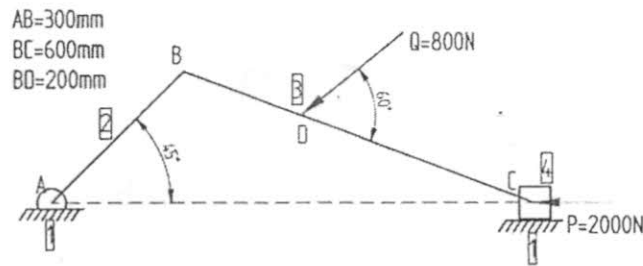


Fig.Q2(b).

3. a) Define and classify kinematic pairs based on the type of relative motion.
- b) What is a kinematic chain? How is it different from a mechanism?
- c) Derive an expression for the balancing of a different rotating mass in the same plane.

Unit – II

4. a) Fig.Q4(a) shows the crank of a slider crank mechanism rotates clockwise at a constant speed of 300 r.p.m. The crank is 150 mm and the connecting rod is 600 mm long. Determine: 1. linear velocity and acceleration of the midpoint of the connecting rod, and 2. angular velocity and angular acceleration of the connecting rod, at a crank angle of 45° from inner dead center position.

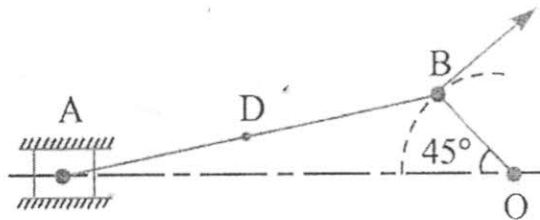


Fig.Q4(a).

- b) Derive the equation for the gyroscopic couple in mechanical systems.
5. a) Derive the equation for the stability of an automobile, considering the condition that no wheel is lifted off the ground while taking a turn in a four-wheeled vehicle.
- b) Find the angle of inclination with respect to the vertical of a two-wheeler negotiating a turn. Given: combined mass of the vehicle with its rider 250 kg; moment of inertia of the engine flywheel 0.3 kg-m^2 ; moment of inertia of each road wheel 1 kg-m^2 ; speed of engine flywheel 5 times that of road wheels and in the same direction; height of centre of gravity of rider with vehicle 0.6m; two-wheeler speed 90 km/h; wheel radius 300 mm; radius of turn 50 m.

6. a) PQRS is a four-bar chain with link PS fixed. The lengths of the links are $PQ = 62.5 \text{ mm}$; $QR = 175 \text{ mm}$; $RS = 112.5 \text{ mm}$; and $PS = 200 \text{ mm}$. The crank PQ rotates at 10 rad/s clockwise. Draw the velocity and acceleration diagram when angle $QPS = 60^\circ$ and Q and R lie on the same side of PS. Find the angular velocity and angular acceleration of links QR and RS.

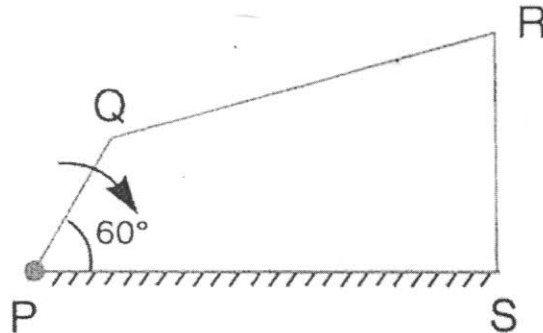


Fig.Q6(a).

- b) Discuss the effect of gyroscopic couple on aeroplane system.

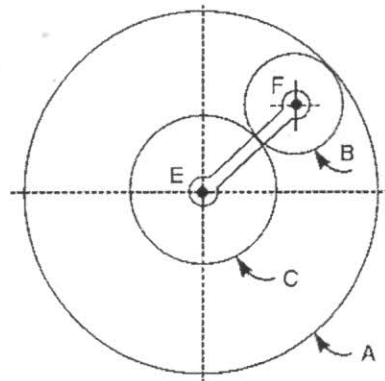
12	L3	3	2
04	L1	3	1

Unit – III

7. a) The number of teeth on each of the two equal spur gears in mesh are 40. The teeth have 20° involute profile, and the module is 6 mm. If the arc of contact is 1.75 times the circular pitch, find the addendum.
b) Derive an expression for the length of arc of contact in a pair of meshed spur gear.
8. a) An epicyclic gear consists of three gears A, B and C as shown in Fig. Q8(a). The gear A has 72 internal teeth and gear C has 32 external teeth. The gear B meshes with both A and C and is carried on an arm EF which rotates about the centre of A at 18 rpm. If the gear A is fixed, determine the speed of gears B and C.

08	L3	4	2
08	L1	5	1

Fig.Q8(a)



- b) Determine the natural frequency of the simple pendulum by neglecting the mass of rod.

12	L3	4	2
04	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
